

TEJASWI SREERANGAM

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EDUCATION

University of Illinois Chicago

Master of Science, Electrical and Computer Engineering

Aug 2024 - May 2026

Savitribai Phule Pune University

Bachelor of Engineering, Electronics and Telecommunications

Jul 2019 - May 2023

PROFESSIONAL EXPERIENCE

InnoIndustry

Volunteer Hardware Engineer

Aug 2026 - Present

United States

- Verify 9 RTL blocks across 2 FPGA/RTL projects using SystemVerilog/UVM, validating reset, control, interface, and datapath behavior against MATLAB golden models.
- Debug RTL/golden-model mismatches through waveform analysis and regression triage, isolating timing, reset, fixed-point, and interface issues while driving coverage toward 95%.
- Collaborate with RTL and algorithm engineers across 9 modules to reproduce failures, close verification gaps, and support module-level integration.

Johnson Controls

Graduate Design Engineer

Oct 2023 - Jun 2024

India

- Performed HW/SW integration and functional validation across 20 multi-story deployments, verifying embedded sensors, controllers, and interfaces against system requirements.
- Investigated 200+ integration records, reproducing failures and tracing interface/configuration issues to root cause, reducing rework by 15%.
- Automated an 8-field engineering traceability workflow using Python, reducing technical-record lookup time by 25% and supporting validation reviews.

PROJECTS

Bare-Metal Bootloader & Firmware Validation

Dec 2025 - Feb 2026

- Implemented a 2-stage STM32F4 bootloader with CRC32 validation, flash partitioning, UART firmware transfer, and Python host tooling.
- Debugged 3 low-level failure areas using JTAG/GDB and validated firmware recovery across 4 scenarios, including corrupted images, interrupted transfers, and rollback.
- Executed 12+ functional and negative tests across boot, image validation, UART transfer, and recovery paths, verifying expected pass/fail behavior and flash integrity.

Embedded Linux Device Driver & Userspace Validation

Sep 2025 - Nov 2025

- Developed an I2C Linux kernel driver supporting 4 core behaviors: probe/remove, sysfs access, and interrupt handling, with a Python userspace validation utility.
- Executed repeated lifecycle and access testing, debugging 3 failure classes, null checks, cleanup, and timing, through dmesg/log analysis.
- Validated 4 driver interfaces/behaviors through repeated probe/remove, sysfs, interrupt, and error-path testing, confirming stable kernel/userspace operation.

CAN Bus Sensor Node & Protocol Validation

Feb 2025 - Apr 2025

- Built and validated a 2-node STM32 CAN network across 3 baud rates (125/250/500 kbps), verifying arbitration, ACK handling, and bus-off recovery.
- Executed regression testing at 500 kbps across 3 injected faults, using logic-analyzer decoding to document error-counter and recovery behavior.
- Captured and analyzed 30+ CAN transactions across nominal and fault-injection tests, verifying arbitration, error handling, and recovery against expected protocol behavior.

SKILLS

- **Hardware & System Validation:** HW/SW integration, Board-level validation, Functional validation, Test execution, Regression testing, Root-cause analysis, Register-level debugging, Failure reproduction, Fault Injection
- **Debug & Lab:** JTAG, GDB, Logic analyzer, CAN protocol decoding, Waveform debugging, dmesg/log analysis
- **Programming & Automation:** Python, C, C++, Bash, Linux, MATLAB, Git
- **Embedded Systems & Interfaces:** ARM Cortex-M4/M7, FreeRTOS, interrupts, DMA, MMIO, UART, SPI, I2C, CAN
- **RTL, Verification & EDA:** Verilog, SystemVerilog, UVM, Functional coverage, Testbench Development, FSMs, Clock/Reset logic, Datapaths, Vivado/XSim, Verilator, GTKWave